Today's Study Questions & Answers

Easy-to-remember, exam-friendly notes — Python & Backend Basics

1. How to take input in List, Tuple and Set?

List: A list uses **[]**.

```
numbers = list(map(int, input("Enter numbers: ").split()))
print(numbers)

Input: 10 20 30 40
Output: [10, 20, 30, 40]
```

Remember: input → split → map → list

Tuple: A tuple uses **()**.

```
numbers = tuple(map(int, input("Enter numbers: ").split()))
print(numbers)

Input: 10 20 30 40
Output: (10, 20, 30, 40)
```

Set: A set is created using **set()**.

```
numbers = set(map(int, input("Enter numbers: ").split()))
print(numbers)

Input: 10 20 30 20 10
Output: {10, 20, 30}
```

Important: A set automatically removes duplicate values.

One formula: list(map(int, input().split()))
tuple(map(int, input().split()))
set(map(int, input().split()))

2. What are the types of API?

API stands for **Application Programming Interface**. It allows different software/applications to communicate with each other.

1. Public API: Available for everyone to use. Example: Google Maps API.

2. Private API: Used inside an organization, such as an internal employee system.

3. Partner API: Available only to selected business partners.

4. Composite API: Combines multiple API requests into one.

Easy trick: P-P-P-C
Public = Everyone
Private = Internal
Partner = Selected partners
Composite = Combination of APIs

Note: REST, SOAP, GraphQL and gRPC are commonly discussed as API architectures/protocol styles.

3. What is Middleware?

Middleware is a function that runs **between the request and the response**.

Think of it like a **security guard** who checks a request before it reaches the next step.

```
app.use((req, res, next) => {
    console.log("Request received");
    next();
});
```

Uses: authentication, permissions, logging, error handling, and modifying requests/responses.

Remember: Middleware = Middle person between Request and Response.

4. What is an .env file?

A **.env file** stores environment variables, configuration values, and secret information.

```
PORT=5000
DATABASE_URL=mongodb://localhost/mydb
API_KEY=123456
```

These values can be accessed in Node.js using **process.env**.

```
console.log(process.env.PORT);
```

Remember: .env = Environment settings + Secrets.

If it contains secrets, it should normally not be uploaded to GitHub.

5. What is a Route?

A **route** defines what should happen when a particular URL and HTTP method are requested.

```
app.get("/users", (req, res) => {
    res.send("All users");
});
```

Here, **GET** is the HTTP method, **/users** is the route, and **res.send()** sends the response.

Common routes: GET /users, POST /users, PUT /users/1, DELETE /users/1

Remember: Route = URL + HTTP method + action.

6. What is a Server File?

A **server file** is the file used to create, configure, and start the backend server.

It is often named **server.js** or **app.js**.

```
const express = require("express");

const app = express();

app.use(express.json());

app.get("/", (req, res) => {
    res.send("Server is working!");
});

app.listen(5000, () => {
    console.log("Server started on port 5000");
```

```
});
```

Remember: Server file = creates + configures + starts the backend server.

7. How to pass a default value in pop() of a dictionary?

The dictionary **pop()** method can take a default value as its second argument.

```
dictionary.pop(key, default_value)
```

```
user = {
    "name": "Ujjawal",
    "age": 20
}

result = user.pop("email", "Not Available")

print(result)
print(user)

Output:
Not Available
{'name': 'Ujjawal', 'age': 20}
```

If the key exists, pop() removes and returns its actual value. If the key does not exist, it returns the default value.

```
result = user.pop("age", "Not Available")
print(result)

Output:
20
```

Remember: pop(key, default) → key exists: remove & return value; key doesn't exist: return default.

Without a default: user.pop("email") raises a KeyError if "email" does not exist.

Quick Revision

List input: list(map(int, input().split()))
Tuple input: tuple(map(int, input().split()))
Set input: set(map(int, input().split()))
Middleware: Between request and response
.env: Environment variables and secrets
Route: URL + HTTP method + action
Server file: Creates, configures and starts server
dict.pop(): pop(key, default)